

Assignment 5

Part A: Authentication Log

Exercise 1: To print

Write sed commands to:

- Print the first log entry.

sed -n '1p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
```

- Print the fifth log entry.

sed -n '5p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
```

- Print entries from lines 2 to 6.

sed -n '2,6p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

- Print the last entry.

sed -n '\$p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Print every second log entry starting from line 1.

sed -n '1~2p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Print all entries containing Failed password.

sed -n '/Failed password/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Print all successful login entries containing Accepted password.

sed -n '/Accepted password/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
```

- Print entries from the first failed login to the authentication failure.

sed -n '/Failed password/,/authentication failure/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 2: Filtering Security Events

Write commands to:

- Display failed login attempts.

sed -n '/Failed password/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Display successful login attempts.

sed -n '/Accepted password/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
```

- Display entries associated with the root account.

sed -n '/root/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Display entries containing Invalid user.

sed -n '/Invalid user/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
```

- Display all events related to IP address 172.16.0.8.

sed -n '/172.16.0.8/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/172.16.0.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

- Display all events that do not contain Accepted password.

sed -n '/Accepted password/!p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Display all events except those generated by sudo.

sed -n '/sudo/!p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Display only the entries that contain either Failed password or Invalid user using multiple -e commands.

sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

- Deleting successful login entries.

sed '/Accepted password/d' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Deleting entries containing Connection closed.

sed '/Connection closed/d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Deleting the first three entries.

sed '1,3d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Deleting entries from lines 4 to 7.

sed '4,7d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Deleting all entries associated with 192.168.1.15.

sed '/192.168.1.15/d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '/192.168.1.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Deleting every second entry starting from the first line.

sed '1~2d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Deleting all events that do not contain Failed password.

sed '/Failed password/!d' auth.log

```
(praneetha@Praneetha)~[/Desktop/Assignment_5_Lab]
$ sed '/Failed password/!d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

Exercise 4: Classifying Security Events

- Replace Accepted password with [SUCCESSFUL LOGIN].

sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace Failed password with [FAILED LOGIN].

sed 's/Failed password/[FAILED LOGIN]/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace Invalid user with [INVALID USER].

sed 's/Invalid user/[INVALID USER]/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/Invalid user/[INVALID USER]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace authentication failure with [AUTHENTICATION FAILURE].

sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace root with PRIVILEGED_USER.

sed 's/root/PRIVILEGED_USER/g' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/root/PRIVILEGED_USER/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for PRIVILEGED_USER from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace every occurrence of server with authentication-server.

sed 's/server/authentication-server/g' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/server/authentication-server/g' auth.log
Sep 10 09:10:12 authentication-server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 authentication-server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 authentication-server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 authentication-server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 authentication-server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 authentication-server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 authentication-server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

- Replace IP address 10.0.0.25 with X.X.X.X.

sed 's/10\.\0\.\0\.\25/X.X.X.X/g' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/10\.\0\.\0\.\25/X.X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.X
```

- Replace 172.16.0.8 with SUSPICIOUS_IP.

sed 's/172\.\16\.\0\.\8/SUSPICIOUS_IP/g' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/172\.\16\.\0\.\8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace username admin with USER_A.

sed 's/admin/USER_A/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/admin/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace username student with USER_B.

sed 's/student/USER_B/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/student/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Replace all digits with X.

sed 's/[0-9]/X/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX
```

- Replace periods in IP addresses with hyphens.

sed -E 's/([0-9]+)\.([0-9]+)\.([0-9]+)\.([0-9]+)/\1-\2-\3-\4/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed -E 's/([0-9]+)\.([0-9]+)\.([0-9]+)\.([0-9]+)/\1-\2-\3-\4/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25
```

- Replace all IP addresses in the sample file with [REDACTED_IP].

sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log

```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

Part B: Web-Server Log

Exercise 6:

Use access.log to solve the below questions

Write sed commands to:

- Print all requests that returned status code 200.

sed -n '/200/p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/200/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200
```

- Print all requests that returned status code 401.

sed -n '/401/p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/401/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

- Print all requests that returned status code 403.

sed -n '/403/p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/403/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
```

- Print all requests to /admin/login.

sed -n '/admin/login/p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/admin/login/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

- Print all POST requests.

sed -n '/POST/p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
```

- Print requests containing ../.

sed -n '/\.\.\//p' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/\.\.\//p' access.log
10.0.0.30 - GET ../../etc/passwd 403
```

- **Print requests containing `<script>`.**

```
sed -n '/<script>/p' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403
```

- **Print requests containing OR.**

```
sed -n '/OR/p' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
```

- **Print all requests except successful requests.**

```
sed -n '/200/!p' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/200!/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
```

- **Print all requests from 10.0.0.25.**

```
sed -n '/^10\.0\.0\.25 /p' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/^10\.0\.0\.25 /p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

- Replace status code 200 with **[ALLOWED]**.

```
sed 's/ 200/ [ALLOWED]/' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/ 200/ [ALLOWED]/' access.log
192.168.1.10 - GET /index.html [ALLOWED]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html [ALLOWED]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html [ALLOWED]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]
```

- **Replace status code 401 with [UNAUTHORIZED].**

```
sed 's/ 401/ [UNAUTHORIZED]/' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/ 401/ [UNAUTHORIZED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

- Replace status code 403 with [BLOCKED].

sed 's/ 403/ [BLOCKED]/' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/ 403/ [BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
```

- Replace /admin/login with /RESTRICTED_RESOURCE.

sed 's/\admin\login\/RESTRICTED_RESOURCE/g' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/\admin\login\/RESTRICTED_RESOURCE/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

- Replace GET with READ_REQUEST.

sed 's/GET/READ_REQUEST/g' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/GET/READ_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

- Replace POST with SUBMIT_REQUEST.

sed 's/GET/READ_REQUEST/g' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

Exercise 8: Grouping Security Operations

- Use { } to perform multiple commands.

For entries containing status code 401:

Replace 401 with [UNAUTHORIZED].

Print the modified line.

```
sed -n '/ 401/ { s/401/[UNAUTHORIZED]/; p; }' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n '/ 401/ { s/401/[UNAUTHORIZED]/; p; }' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
```

- For entries containing status code 403:

Replace 403 with [BLOCKED].

Replace the source IP address with [REDACTED_IP].

Print the modified entry.

```
sed -n -E '/ 403/ { s/403/[BLOCKED]/; s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; p; }' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n -E '/ 403/ { s/403/[BLOCKED]/; s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; p; }' access.log
[REDACTED_IP] - GET /search?id=1'OR'1'=1 [BLOCKED]
[REDACTED_IP] - GET ../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=<script> [BLOCKED]
```

- For requests to /admin/login:

Replace the URL with /PROTECTED_PAGE.

Replace the source IP with [REDACTED_IP].

Print the result.

```
sed -n -E '/\admin\login/ { s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; s/\admin\login\/PROTECTED_PAGE/g; p; }' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -n -E '/\admin\login/ { s/^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/; s/\admin\login\/PROTECTED_PAGE/g; p; }' access.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401
```

Exercise 9: Adding Incident Labels

- Insert the following heading before the first line of auth.log:

SECURITY AUTHENTICATION LOG

```
sed '1i SECURITY AUTHENTICATION LOG' auth.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 'i SECURITY AUTHENTICATION LOG' auth.log
SECURITY AUTHENTICATION LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Append the following message after the final entry:

END OF SECURITY LOG

sed '\$a END OF SECURITY LOG' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '$a END OF SECURITY LOG' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
END OF SECURITY LOG
```

- Insert [WARNING] before every failed login.

sed '/Failed password/i [WARNING]' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/Failed password/i [WARNING]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
[WARNING]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
[WARNING]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
[WARNING]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Append Review this event after every invalid-user entry.

sed '/Invalid user/a Review this event' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/Invalid user/a Review this event' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Review this event
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.

sed '/ 403/i [SUSPICIOUS WEB REQUEST]' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/ 403/i [SUSPICIOUS WEB REQUEST]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
[SUSPICIOUS WEB REQUEST]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[SUSPICIOUS WEB REQUEST]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

- Append [REQUIRES INVESTIGATION] after every 401 entry.

sed '/ 401/a [REQUIRES INVESTIGATION]' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/ 401/a [REQUIRES INVESTIGATION]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
[REQUIRES INVESTIGATION]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[REQUIRES INVESTIGATION]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

- Replace all entries containing Connection closed with:

[CONNECTION TERMINATED]

sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/Connection closed/c [CONNECTION TERMINATED]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]
```

- Replace all successful web requests with:

[NORMAL WEB REQUEST]

sed '/ 200/c [NORMAL WEB REQUEST]' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/ 200/c [NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
```

Exercise 10: Separating Log Entries with G

Use G to:

- Add a blank line after every authentication event.

sed 'G' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15

Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Add a blank line after every failed login.

sed '/Failed password/G' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Add a blank line after every blocked web request.

sed '/ 403/G' access.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '/ 403/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403

192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403

172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403

192.168.1.25 - GET /about.html 200
```

- Add a blank line after every second entry starting with the first entry.

sed '1~2G' auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed '1~2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 11:

- Write failed logins to failed_logins.txt.

sed -n '/Failed password/p' auth.log > failed_logins.txt

- Write successful logins to successful_logins.txt.

sed -n '/Accepted password/p' auth.log > successful_logins.txt

- Write invalid-user events to invalid_users.txt.

sed -n '/Invalid user/p' auth.log > invalid_users.txt

- Write 401 web events to unauthorized_requests.txt.

sed -n '/ 401\$/p' access.log > unauthorized_requests.txt

- Write 403 web events to blocked_requests.txt.

sed -n '/ 403\$/p' access.log > blocked_requests.txt

- Write all events related to 172.16.0.8 to suspicious_ip.txt.

sed -n '/172.16.0.8/p' auth.log > suspicious_ip.txt

Exercise 12:

- Label successful and failed logins.

```
sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' \-e 's/Failed password/[FAILED LOGIN]/g' auth.log
```

```
(praneetha@Praneetha)~[~/Desktop/Assignment_5_Lab]
$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/g' \-e 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

- Label 401 and 403 web requests.

```
sed -e 's/401/[UNAUTHORIZED]/g' \-e 's/403/[BLOCKED]/g' access.log
```

```
(praneetha@Praneetha)~[~/Desktop/Assignment_5_Lab]
$ sed -e 's/401/[UNAUTHORIZED]/g' \-e 's/403/[BLOCKED]/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
```

- Delete successful logins and print failed logins.

```
sed -n -e '/Accepted password/d' -e '/Failed password/p' auth.log
```

```
(praneetha@Praneetha)~[~/Desktop/Assignment_5_Lab]
$ sed -n -e '/Accepted password/d' -e '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

- Replace root with [PRIVILEGED_USER] and mask all IP addresses.

```
sed -E -e 's/root/[PRIVILEGED_USER]/g' \-e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

```
(praneetha@Praneetha)~[~/Desktop/Assignment_5_Lab]
$ sed -E -e 's/root/[PRIVILEGED_USER]/g' \-e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

- Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.

```
sed -e 's/GET/READ_REQUEST/g' \-e 's/POST/SUBMIT_REQUEST/g' access.log
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -e 's/GET/READ_REQUEST/g' \-e 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

Exercise 13:

Create security_report.sed containing commands to:

Insert SECURITY INCIDENT REPORT before the first line.

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER ATTEMPT].

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace all sample IP addresses with [REDACTED_IP].

Append END OF INCIDENT REPORT after the last line.

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ nano security_report.sed
```

security_report.sed:

```
1i SECURITY INCIDENT REPORT

s/Failed password/[FAILED LOGIN]/g

s/Invalid user/[INVALID USER ATTEMPT]/g

s/Accepted password/[SUCCESSFUL LOGIN]/g

s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g

$a END OF INCIDENT REPORT
```

Execute it using:

sed -f security_report.sed auth.log

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
```

Save the output:

sed -f security_report.sed auth.log > incident_report.txt

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
```

Exercise 14: Modifying a Security Configuration File

- Create security.conf:

PermitRootLogin yes

PasswordAuthentication yes

MaxAuthTries 6

FirewallStatus disabled

LogLevel INFO

GuestLogin enabled

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ nano security.conf
```

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ cat security.conf
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

- First make a backup:

cp security.conf security.conf.backup

```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]
$ cp security.conf security.conf.backup
```

- Use sed -i to:

Change PermitRootLogin yes to PermitRootLogin no.

sed -i 's/PermitRootLogin yes/PermitRootLogin no/' security.conf

Change MaxAuthTries 6 to MaxAuthTries 3.

sed -i 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf

Change FirewallStatus disabled to FirewallStatus enabled.

```
sed -i 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf
```

Change LogLevel INFO to LogLevel VERBOSE.

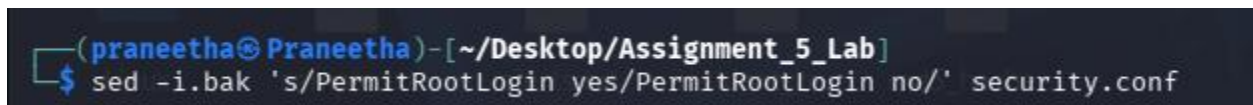
```
sed -i 's/LogLevel INFO/LogLevel VERBOSE/' security.conf
```

Change GuestLogin enabled to GuestLogin disabled.

```
sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf
```

- Use the safer backup form:

```
sed -i.bak 'sed-command' security.conf
```

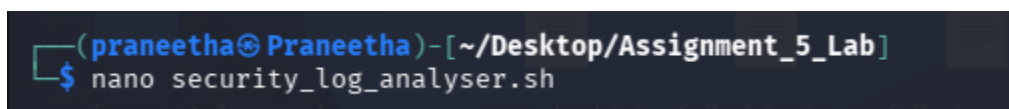


```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]  
$ sed -i.bak 's/PermitRootLogin yes/PermitRootLogin no/' security.conf
```

This exercise demonstrates configuration editing only; it does not configure the actual system.

Exercise 15:

- Write a Bash script named security_log_analyser.sh.



```
(praneetha@Praneetha)-[~/Desktop/Assignment_5_Lab]  
$ nano security_log_analyser.sh
```

- The script should:

Accept a log filename as a command-line argument.

Check whether the argument is provided.

Check whether the file exists.

Extract failed login attempts.

Extract invalid-user attempts.

Replace sensitive IP addresses with [REDACTED_IP].

Add the heading SECURITY ALERT REPORT.

Add [ALERT] before suspicious entries.

Append END OF REPORT.

Save the result in security_alert_report.txt.

Display the generated report.

```
#!/usr/bin/bash

if [ $# -eq 0 ]
then
    echo "Error: Please provide a log filename."
    exit 1
fi

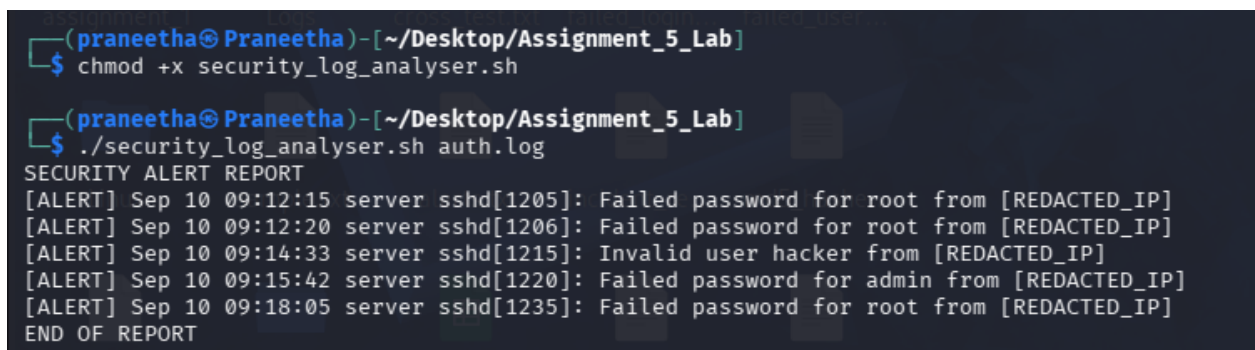
if [ ! -f "$1" ]
then
    echo "Error: File does not exist."
    exit 1
fi

echo "SECURITY ALERT REPORT" > security_alert_report.txt

sed -n \
    -e '/Failed password/p' \
    -e '/Invalid user/p' "$1" |
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' |
sed 's/^[ALERT] //' >> security_alert_report.txt

echo "END OF REPORT" >> security_alert_report.txt

cat security_alert_report.txt
```



```
(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ chmod +x security_log_analyser.sh

(praneetha@Praneetha) - [~/Desktop/Assignment_5_Lab]
$ ./security_log_analyser.sh auth.log
SECURITY ALERT REPORT
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
END OF REPORT
```